

SVEP1 enables efficient binding of Angiopoietin-2 to the TIE1 receptor, allowing receptor phosphorylation and downstream signaling

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eLife Assessment

This manuscript focuses on developing a structural model of how the multidomain ECM protein SVEP1 enables Angiopoietin (ANG) binding to the orphan receptor TIE1, resulting in downstream receptor phosphorylation and signaling. This is a potentially **important** study, however, it currently lacks key controls and is therefore **incomplete**. The data will be of interest to scientists working in vascular biology and RTK signaling.

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Abstract

The molecular mechanisms that drive (lymph-)angiogenesis are crucial to understand diseases, such as lymphedema, that are caused due to malformations of the lymphatic vasculature. Recently an interaction between the secreted protein Svp1, a key regulator in lymphangiogenesis, and the transmembrane receptor Tie1 was shown in zebrafish, human and mice. Here, guided by *in silico* AlphaFold-multimer structure predictions of highly confident SVEP1 complexes, we assert with protein binding studies that the human CCP20 domain is the primary binding site for TIE1, and also show that in addition the CCP5-EGFL7 fragment of SVEP1 engages TIE1 as a secondary site. We further demonstrate that SVEP1 mediates strong binding of ANG2 and TIE1, and that combined stimulation of hDLECs with SVEP1 and ANG2 leads to phosphorylation of TIE1. TIE1 activation by SVEP1 and ANG2 enables downstream signaling and in turn potentiates nuclear exclusion of FOXO1 and phosphorylation of AKT compared to SVEP1 or ANG2 alone. We present a model where ANG1/2 dimers bind to both SVEP1 and TIE1, resulting in multiple TIE1 receptor molecules being recruited to a multimeric complex at the cell membrane, potentially amplifying its signaling capacity.

Introduction

Lymphatic vessels are closely associated to blood vessels, and malformations of the lymphatic vasculature lead to lymphedema, obesity, or chronic inflammatory diseases (Mäkinen et al. 2021 [↗](#); Petrova and Koh 2018 [↗](#)). Next to the well described lymphatic VEGFC/VEGFR3 pathway (Hogan et al. 2009 [↗](#); Bos et al. 2011 [↗](#); Le Guen et al. 2014 [↗](#); Jeltsch et al. 2014 [↗](#); Roukens et al. 2015 [↗](#)), it was shown that Sushi, von Willebrand factor type A, EGF and pentraxin domain containing 1 (SVEP1), also referred to as Polydom, is another key player in lymphangiogenesis (Karpanen et al. 2017 [↗](#); Morooka et al. 2017 [↗](#)). Svp1 is a large extracellular matrix protein which is expressed in mesenchymal cells that functions non-cell-autonomously (Karpanen et al. 2017 [↗](#); Morooka et al. 2017 [↗](#)). Only recently, it was discovered that SVEP1 binds to TIE1 (Hußmann et al. 2023 [↗](#); Sato-Nishiuchi et al. 2023 [↗](#)), and we were able to provide *in vivo* evidence for the genetic interaction of *svp1* and *tie1* in zebrafish. Multiple phenotypic hallmarks of *svp1* mutants are phenocopied by *tie1*, but not *tie2* mutants, including (among others) a reduced number of parachordal lymphangioblasts (PLs) (Hußmann et al. 2023 [↗](#)). Furthermore, a specific aspect of the facial lymphatics in zebrafish, the facial collecting lymphatic vessel (FCLV), develops in a *svp1* and *tie1* dependent, but *vegfc/vegfr3* and *tie2* independent manner (Hußmann et al. 2023 [↗](#); Morooka et al. 2024 [↗](#)).

The discovery of the novel molecular interaction between SVEP1 and TIE1 provided an invitation to further investigate how SVEP1 is linked to the already well-known TIE-ANG pathway. The TIE-ANG pathway triggers a signaling cascade, which is required for lymphatic and blood vessel development, and has a unique role in maintaining vascular stability. Two tyrosine kinase receptors, tyrosine-protein kinase receptor 1 (TIE1) and tyrosine endothelial kinase (TEK), also known as TIE2 (Dumont et al. 1993 [↗](#); Partanen et al. 1992 [↗](#)), are activated through binding of angiopoietin ligands including ANG1 and ANG2 (Davis et al. 1996 [↗](#); Maisonpierre et al. 1997 [↗](#)). TIE2 is considered as the main player in angiogenesis and its phosphorylation, triggered by ANG1 binding, induces downstream signaling via PI3K/Akt, which leads to inhibition of the transcription factor Forkhead box protein O1 (FOXO1) and repression of FOXO1 target genes - such as *Ang2* (Kim et al. 2000; Daly et al. 2004 [↗](#)). Although TIE2 is needed for lymphangiogenesis in mouse embryos, as genetic deletion of *Tie2* in lymphatic vessels leads to subcutaneous edema (Souma et al. 2018 [↗](#)), postnatal knock-out of *Tie2* in lymphatics of newborn mice only has an effect on lymphatic collecting vessels but not on cutaneous lymphatic capillaries (Korhonen et al. 2022 [↗](#); Shen et al. 2014 [↗](#)). While ANG1 is an agonist of TIE2, ANG2 acts as a context-dependent weak agonist or antagonist for TIE2 that can inhibit the ANG1–TIE2 signaling axis (Saharinen, Eklund, and Alitalo 2017 [↗](#)). TIE1 blocks the signaling cascade in a context dependent manner by forming heterodimers with TIE2 (Hansen et al. 2010 [↗](#); Marron et al. 2000 [↗](#); Saharinen et al. 2005 [↗](#); Savant et al. 2015 [↗](#); Seegar et al. 2010 [↗](#)). On the other hand, *Tie1* deletion leads to defects in the lymphatic capillary network and primary lymphatic network remodeling (Korhonen et al. 2022 [↗](#); Shen et al. 2014 [↗](#)).

In the present study we extend our understanding of SVEP1 binding to TIE1 utilizing *in silico*, *in vitro*, and *in vivo* techniques. While the CCP20 module has been suggested to be the only SVEP1 domain that binds TIE1 (Sato-Nishiuchi et al. 2023 [↗](#)), we here introduce another, distal segment of the SVEP1 chain that is also able to bind TIE1 and is able to rescue the lymphatic defects of *svp1* mutants *in vivo*. We further show that the binding of ANG2 to TIE1 is enhanced by SVEP1 leading to phosphorylation and downstream signaling of TIE1, thereby potentiating phosphorylation of AKT and nuclear exclusion of FOXO1. Our study helps to understand the interaction of important signaling pathway members involved in (lymph-)angiogenesis and provides a model that suggests local TIE1 multimerization through bridging functions of ANG1/2 in the presence of SVEP1.

Results

SVEP1 binds to the D1-D2 domain of TIE1

Previously it had been shown that solid-phase binding assays of recombinant full-length murine SVEP1 protein display a significant binding of SVEP1 to TIE1, but not TIE2 (Sato-Nishiuchi et al. 2023 [\[1\]](#)). In a binary setup, i.e., without other proteins being present, we confirmed this observation utilizing Biacore and negative staining assays (**Figure 1** [\[1\]](#)). Furthermore, and since AlphaFold2 (AF2) and AF3 are remarkably accurate predictors of evolutionarily conserved protein-protein interactions, we used their multimer capabilities to search for significant contact points between the full length, 3571 amino acid human SVEP1 chain, and potential binding partners TIE1 and TIE2. As shown by the comparative PAE (Predicted Aligned Error, a prediction quality and model confidence metric) plots of the SVEP1 matches with TIE1 and TIE2 with AF3 (**Figure 1A, B** [\[1\]](#)), we detect an unequivocal, high-affinity interaction between the headpiece or N-terminal D1-D4 domains of TIE1 and the CCP19-21 stretch of SVEP1, in line with the results of Sato-Nishiuchi et al. (2023) [\[1\]](#) which center on the CCP20 domain of murine SVEP1 for TIE1 capture. By contrast, the PAE plot for the D1-D4 segment of TIE2 fails to show an equivalent interaction signal with the CCP19-21 stretch of SVEP1. The more focused interrogation and modeling of equivalent N-terminal TIE1 and TIE2 D1-D4 domains with the isolated CCP19-CCP21 domain fragment of SVEP1 (using both AF2.3 and AF3) affirms the strong binding of SVEP1 to TIE1, positioning the groove between the side-by-side packed Ig-like D1-D2 domains of TIE1 in close contact with the CCP20 module of SVEP1, revealing a buried surface area of 1771 Å² marked by 12 H-bonds and 3 salt bridges by PISA analysis (**Supplementary Figure 1** [\[1\]](#)). The SVEP1-binding domains of TIE1 are buttressed by a Cys-rich D3 domain that links to an outstretched D4 Ig module that further connects to a linear array of D5-D7 Fn3 domains before the hydrophobic TM helix. In this closer examination of molecular fragments of TIE2 and SVEP1, AF3 gauges that the TIE2 headpiece has a less confident and qualitatively weaker binding to the CCP19-21 fragment of SVEP1, and the resulting multimer complex model shows domains D1-D4 of TIE2 perched atop the SVEP1 CCP20 module of SVEP1. While this TIE2 engagement is in a similar pose to TIE1, PISA analysis details a smaller buried surface area of 1482 Å² (with 6 H-bonds and 3 salt bridges) than with TIE1, suggesting a lower affinity engagement. To verify the modelling results, we performed Biacore assays with a 70 kDa version of the SVEP1 protein that contains the CCP20 domain (**Figure 1C** [\[1\]](#)). We observed interaction of SVEP1 with TIE1, but not with TIE2 in the absence of other proteins. Furthermore, utilizing negative staining and transmission electron microscopy, we were able to directly visualize the attachment of SVEP1 and TIE1 (**Figure 1D** [\[1\]](#)).

The CCP6-EGFL7 domain of SVEP1 constitutes an additional binding domain for TIE1

It had previously been published that the CCP20 domain of murine SVEP1 is responsible for the binding to TIE1 (Sato-Nishiuchi et al. 2023 [\[1\]](#)), a finding which is in line with our data using human versions of the respective proteins (**Figure 1** [\[1\]](#)). In an attempt to determine whether there are additional SVEP1 sites that can bind Tie1, we performed *in vivo* rescue experiments using different deletion constructs of murine SVEP1. Injecting the full length *Svep1* construct (kindly provided by K. Sekiguchi) led to a significant reduction of the lymphatic defects in the trunk of *svep1* mutants compared to non-injected controls. The N-terminus appeared to be of low functional importance, as the versions PolC, ΔN-EGFL6, and ΔN-PTX of SVEP1 rescued the phenotype of *svep1* mutants (**Figure 2A** [\[1\]](#) and **Supplementary Figure 2** [\[1\]](#)). Therefore, we hypothesized that the C-terminus (ΔN-PTX), consisting of a significant stretch of CCP elements and four EGF domains, is required for SVEP1 function during lymphangiogenesis. To further test which domains are able to rescue the phenotype, we generated additional deletion constructs based on ΔN-PTX. Deletion of the C-

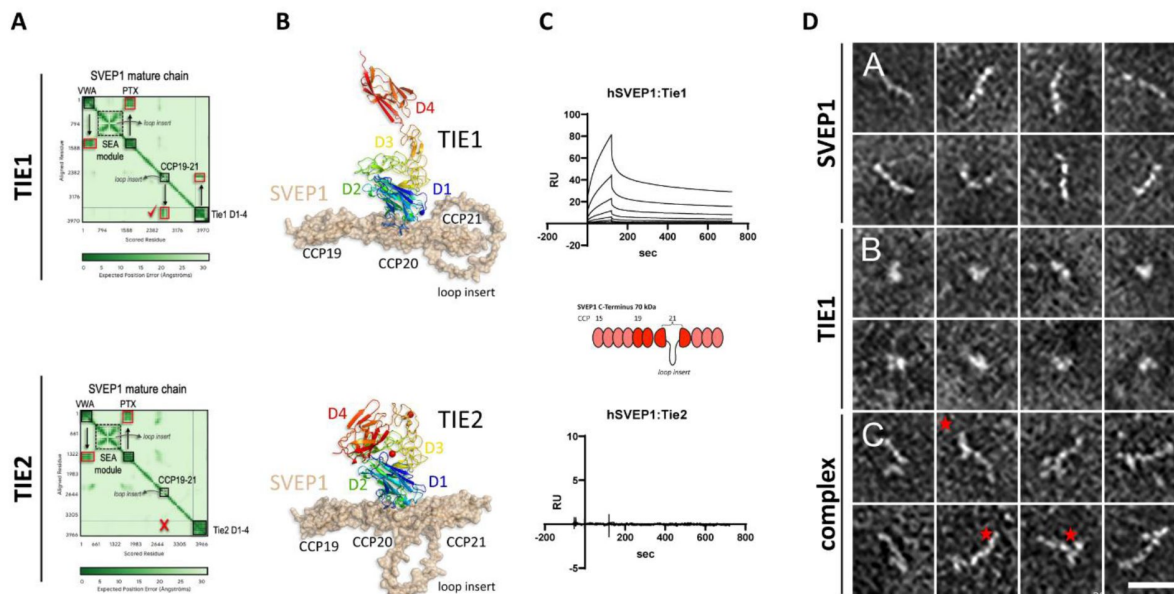


Figure 1

The CCP20 domain of SVEP1 contacts the D1-D2 headpiece domains of TIE1.

(A) Comparative PAE plots of full-chain SVEP1 displays a primary and high-confidence binding site for TIE1 (domains D1-D4) by AF3-multimer modeling, that localizes to the CCP19-21 stretch, notably encompassing CCP20 that has a sizable 45 amino acids insert loop. The PAE plot for TIE2 D1-D4 does not show an equivalent interaction signal for the same CCP19-21 region. (B) A discrete AF3 model of TIE1 bound to CCP19-21 shows that the Ig-like D1-D2 headpiece domains of TIE1 (linked by a Cys-rich D3 to an 'open' or flared-out D4 Ig module) are in contact with CCP20. By contrast, the AF3 model of TIE2's N-terminal domains (that are packed in a more compact fashion, with D4 notably in a 'closed' position) is more loosely perched atop CCP20 with a qualitatively much lower affinity. (C) Quantitative analysis of TIE1-SVEP1 interaction using Biacore assays. In the absence of other proteins, a 70kD version of SVEP1 (depicted in diagram fashion) containing the CCP20 domain binds TIE1, but does not bind TIE2. (D) Shadowing EM shows binding of SVEP1 and TIE1. PAE: Predicted aligned error; AF3: AlphaFold 3; CCP: complement control protein (sushi repeats).

terminus from CCP22 onwards retained significant rescuing activity (**Figure 2** [↗](#), Δ N-PTX Δ CCP22-C). In contrast, deletion from the N-terminal part including CCP21 (Δ N-CCP21) were found to not rescue the phenotype of *svep1* mutants anymore. Therefore, some functionally relevant domains must be located between CCP5 and CCP22. By generating and injecting Δ N-PTX Δ 7-34 (**Supplementary Figure 2** [↗](#)), we still observe a significant rescue and therefore concluded that CCP5-EGF7, which are present in both Δ N-PTX Δ CCP22-C and Δ N-PTX Δ 7-34, must be sufficient to rescue the loss of the TD in *svep1* mutants. To confirm this hypothesis, we also generated a CCP5-EGF7 construct and showed that injection of an mRNA encoding only these three domains (161 amino acids) can provide a significant rescue of the lymphatic defects in the zebrafish trunk of *svep1* mutants.

In order to test if CCP5-EGF7 is not only sufficient, but also essential for Svep1 function, we generated an in-frame deletion of these domains using CRISPR/Cas9 technology in zebrafish. However, we did not observe any lymphatic defects in these mutants (**Figure 2B** [↗](#)), indicating that in this case other CCP domains, most likely CCP20, which is the strongest binding site for Tie1, are able to fulfil the function of Svep1.

Consistent with these findings, we confirmed by AF2.3 modelling that human TIE1 binds to a secondary site in human SVEP1 comprising the CCP5-EGFL7 stretch that is adjacent to the central PTX domain (**Figure 2C** [↗](#)). The same D1-D2 headpiece of TIE1 that recognizes the CCP20 domain of SVEP1 mounts the CCP5-EGFL7 chain by binding across the interface between packed CCP6-EGFL7 modules, specifically atop the EGFL7 Ca^{2+} -coordinating loop. The binding of TIE1 (D1-D2) and CCP6-EGFL7 show a less confident signal by PAE plot, suggesting a weaker interaction, while PISA analysis shows 1748 Å² of buried surface area (including 9 H-bonds and 2 salt bridges) that nevertheless compares to the high-affinity, primary CCP20 binding site (**Figure 2D** [↗](#)). To confirm this prediction we conducted immunoprecipitation assays, where the cell lysate of TIE1 transfected 293T HEK cells was added to different versions of the SVEP1 protein that were bound to beads. We were able to detect that SVEP1 versions containing the CCP20 and CCP21 domain strongly bound TIE1, whereas the version without the CCP21 domain bound much weaker. As a negative control we used, among others, the CCP19-21 version with a mutation that leads to an exchange of two amino acids in the CCP20 domain, and is predicted not to bind TIE1 (Sato-Nishiuchi et al. 2023 [↗](#)). Finally, we tested the binding capacity of CCP5-EGFL7 using the immunoprecipitation assay and could confirm the binding ability to TIE1 (**Figure 2E** [↗](#)). Taken together, we could confirm that the CCP5-EGFL7 domain is able to bind TIE1 and can function *in vivo*. However, CCP5-EGFL7 appears to be a weaker binder compared to CCP20 and is not essential for the function of Svep1 during lymphangiogenesis.

ANG1/2 strengthens the binding capacity of SVEP1 and TIE1

Murine SVEP1 binds ANG1 and ANG2 *in vitro* (Morooka et al. 2017 [↗](#)) and it has long been known that ANG1 and ANG2 bind to TIE2 (Davis et al. 1996 [↗](#); Maisonpierre et al. 1997 [↗](#)). Both AF2 and AF3 multimer models effectively capture TIE1 in a complex with SVEP1 CCP20, so we next explored how the addition of ANG1/2 would affect this assembly (**Figure 3A** [↗](#)). The affinity of TIE1 for SVEP1 is significantly increased by the simultaneous binding of the C-terminal domains of ANG1 or ANG2 to the CCP20 module, as suggested by the heightened confidence levels of the PAE plots for the respective ternary assemblies, that improve on the binary coupling of just TIE1 to SVEP1 (**Suppl. Figure 3A** [↗](#)). By contrast (not shown), TIE2 was not predicted to stably co-bind with ANG1/2 to CCP19-CCP21 of SVEP1 because the weaker binding of TIE2 with CCP19-CCP21 of SVEP1 fails to nucleate the formation of a ternary complex. The conformation of the TIE1-ANG1/2 heteromers in the SVEP1-bound pose resembles the solved X-ray structures of TIE2 bound to either ANG1 or ANG2 (PDB codes 4K0V and 2GY7, respectively, in the absence of any experimental TIE1-ANG complexes) (**Supplementary Figure 3B** [↗](#)), and furthermore, the equivalent ANG1/2-binding epitopes in TIE1 are not blocked by SVEP1 binding. Whereas the two TIE2 X-ray complexes show a modest buried surface area of ~1200 Å² (with 2 H-bonds and 2 salt bridges) against ANG1/2, the addition of the SVEP1-TIE1 contact (~1800 Å², with 4 H-bonds) and the new SVEP1-ANG1 or SVEP1-

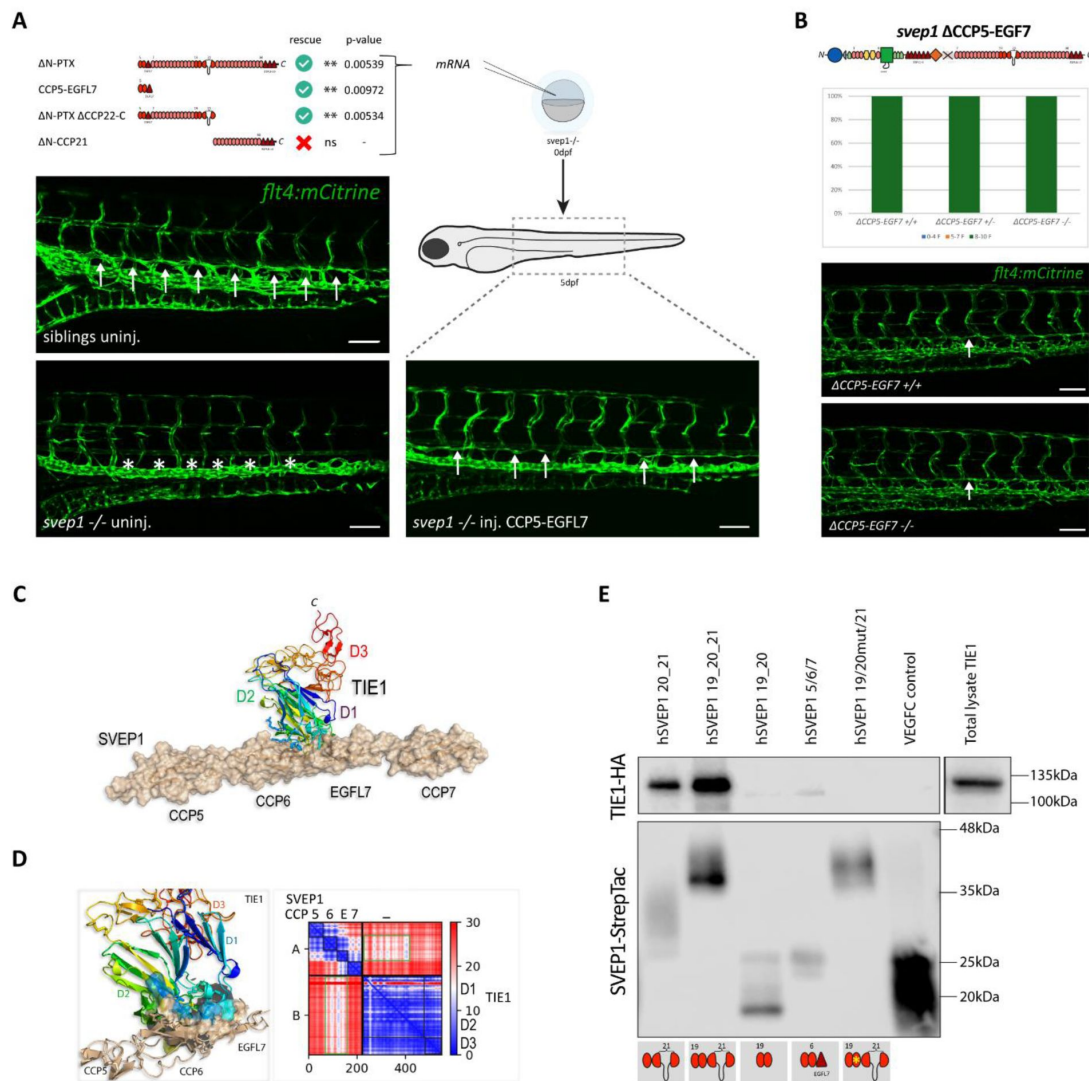


Figure 2

The CCP5-EGF7 domain of SVEP1 constitutes an additional binding domain for TIE1 which is sufficient, but not required for the lymphatic function of Svep1.

(A) Injection of different versions of murine *Svep1* rescue constructs in *svep1* ^{-/-} zebrafish zygotes at single cell stage was followed by quantifying the number of TD fragments at 5 dpf. Depicted in the confocal images, siblings show a complete TD, indicated by arrows. Non-injected *svep1* mutants show a loss of the TD, indicated by asterisks, while mutants injected with the CCP5-EGF7 mRNA have significant more fragments of TD compared to the control (*flt4:mCitrine* transgene). For statistical analysis, the number of TD fragments of non-injected *svep1* mutants was compared to the injected mutants (Mann-Whitney test). Scale bar = 100 μm. (B) Using CRISPR/Cas9 technology, an in-frame deletion of CCP5-EGF7 was generated in zebrafish. Homozygous mutants (ΔCCP5-EGF7) show no lymphatic defects (*svep1*ΔCCP5-EGF7 +/+ n=22, *svep1*ΔCCP5-EGF7 +/- n=53, *svep1*ΔCCP5-EGF7 -/- n=21). Therefore, the region is not required for the lymphatic function of Svep1. Scale bar =100μm. (C) The AF3 model predicts that TIE1 has a secondary binding site in the SVEP1 chain in the CCP5-7 stretch (that closely follows the PTX domain) that bridges CCP6 and (Ca²⁺-binding) EGFL7 modules. The same D1-D2 headpiece domains of TIE1 that recognize CCP20 are employed. In the present AF2.3 model, with the outward-facing D4 domain in TIE1 being omitted, since it does not participate in binding. (D) The binding of TIE1 domains D1-D2 with the CCP6-EGFL7 unit show a confident, but moderate affinity signal in the PAE plot, and the PISA-analyzed EGFL7 domain of SVEP1 alone is able to bind TIE1. Different versions of human SVEP1-StrepTag were immunoprecipitated and associated TIE1 was detected via Western blot analysis. In three of four replicates, the CCP5-EGFL7 bound to TIE1 much weaker in comparison to the SVEP1 proteins that contain the CCP20 domain. TD, thoracic duct; dpf, days post fertilization PAE: Predicted aligned error; AF3: AlphaFold 3; CCP: complement control protein (sushi repeats).

ANG2 interface (831 Å² and 11 H-bonds/3 salt bridges, or 949 Å² and 13 H-bonds/4 salt bridges, respectively), create an overwhelmingly more extensive interface— approximately 3900 Å² in total, with a number of further discriminating H-bonds and salt bridges— than the binary TIE1-ANG1/2 complex alone. For this reason, SVEP1's CCP20 module acts as a critical binding partner for TIE1, but not TIE2, when interacting with ANG1/2.

Co-transfection of either ANG1 or ANG2 together with TIE1 and subsequent immunoprecipitation of SVEP1 bound to beads verified the model prediction (**Figure 3B, C**). Whereas SVEP1 levels in all samples were similar, the level of TIE1 bound to SVEP1 significantly increased when ANG1/2 was present in the lysate. TIE1 co-transfected with ANG1 led to a slight, but not significant increase of TIE1 levels.

Stimulation by SVEP1 leads to phosphorylation of TIE1, nuclear exclusion of FOXO1 and phosphorylation of AKT

In previous studies, phosphorylation of TIE1 could not be detected after stimulation of cells with SVEP1 (Sato-Nishiuchi et al. 2023) or ANG2 (Saharinen et al. 2005) only, whereas stimulation of EA.hy926 (Saharinen et al. 2005). Our aim was to test whether cooperative binding of SVEP1 and angiopoietins to TIE1 can lead to TIE1 phosphorylation, where SVEP1 constitutes the connecting link between TIE1 and ANG2. Indeed, after stimulation of hdLECs with both SVEP1 and ANG2, phosphorylation of TIE1 at residue pY¹⁰⁰⁷ was detected (**Figure 4A** and **D**). Also, the total pTyr level (detected by 4G10 antibody) of TIE1 was increased. To further test whether SVEP1-ANG2-stimulation leads to downstream signaling, we analyzed pAKT levels. Stimulation with either ANG2 or SVEP1 led to phosphorylation of AKT which was amplified by simultaneous stimulation with ANG2 and SVEP1. The effect of SVEP1 stimulation on AKT signaling could be blocked by an ANG2 blocking antibody suggesting that SVEP1 does not induce signaling on its own, but enables signaling of ANG2 (**Figure 4B** and **E**). Knock-down of either TIE1 or TIE2 could reduce activation of pAKT, leading to the assumption that both, TIE1 and TIE2 are necessary for the phosphorylation of AKT after SVEP1-ANG2 stimulation and thus downstream signaling. Furthermore, we analyzed FOXO1 nuclear exclusion, a downstream event of the PI3K/AKT pathway, by stimulating hdLECs with either ANG2 or SVEP1, or both proteins in combination, and testing for nuclear versus cytoplasmic FOXO1 ratios. It has already been shown that SVEP1 activates the PI3K-Akt signaling pathway *in vitro*, with nuclear exclusion of FOXO1 (Sato-Nishiuchi et al. 2023). We observed enhanced nuclear exclusion of FOXO1 in hdLECs stimulated with combined SVEP1 and ANG2 compared to ANG2 or SVEP1 alone (**Figure 4C**), indicating that ANG2 potentiates the activation of PI3K-Akt and thus inactivates FOXO1. ANG2 by itself had a comparable effect on the FOXO1 inactivation as SVEP1, with both proteins showing a significant effect (**Figure 4F**).

SVEP1, TIE1 and ANG1/2 are predicted to form higher order complexes

The results show that SVEP1 together with ANG2 lead to phosphorylation of TIE1 which then activates the PI3K-AKT signaling pathway leading to nuclear exclusion of FOXO1. Utilizing AF3 modeling we were able to confidently predict a stable 2:2:2 complex of SVEP1, TIE1 and ANG1/2, where the previously modeled, high affinity ternary complexes, now are incorporated into a larger assembly of dimerized TIE1 receptors (**Figure 5A**). Focusing on the main interaction hub, the converging TIE1 and TIE1' chains show domain-swapped D4 modules in the previously seen 'open' conformation, with D1-D2 headpiece domains firmly affixed to both SVEP1 CCP20 and ANG1 or ANG2 (**Figure 5B**). Dimeric ANG1/2 molecules with their C-terminal binding domains attached to short linkers, can simultaneously bind different SVEP1 chains in a neighboring – and not the same – interaction hub, by cross-lacing of their N-terminal coiled-coil dimers that brings several TIE1 receptors into spatial proximity on the cell surface. Both TIE1 and TIE2 receptors are

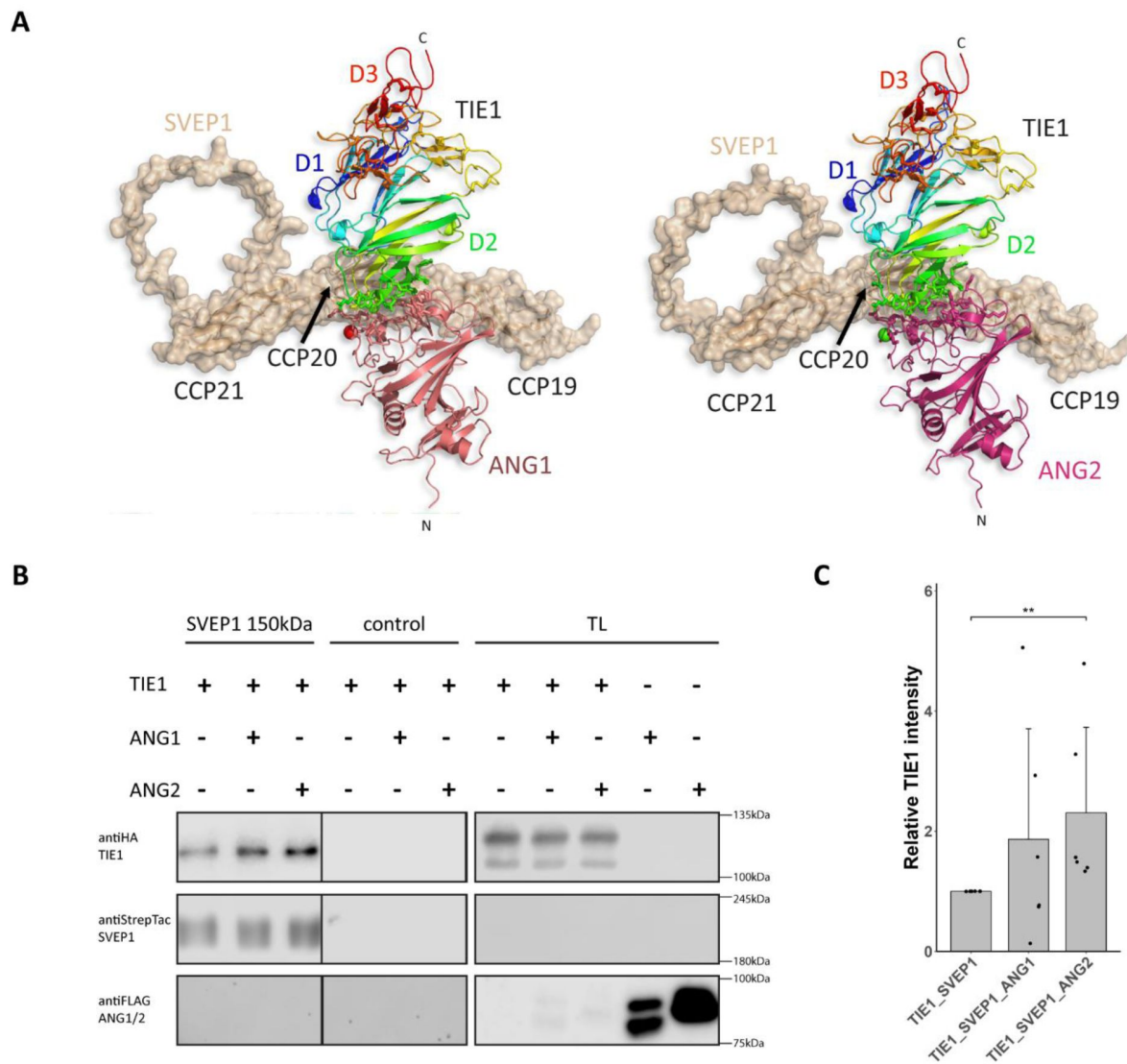


Figure 3

ANG1/2 strengthens the binding capacity of SVEP1 and TIE1.

(A) AF3-modeling of SVEP1 CCP19-21 domain and TIE1 together with ANG1/2. The affinity of TIE1 for SVEP1 is considerably heightened by the co-binding of ANG1 or ANG2 C-terminal domains to the CCP20 module. (B) Co-immunoprecipitation of human TIE1 co-transfected with either human ANG1 or ANG2 in 293T HEK cells. Human 150kDa SVEP1-StrepTac was immunoprecipitated and associated TIE1 and ANG1/2 were detected via Western blot analysis. (C) Quantification of the TIE1 levels with SVEP1 levels as a reference confirmed a significant increase in binding affinity of SVEP1 and TIE1 together with ANG2 (**p.adj = 0.008; Mann-Whitney U test; Values are presented as means ± SD; individual data points for each experiment).

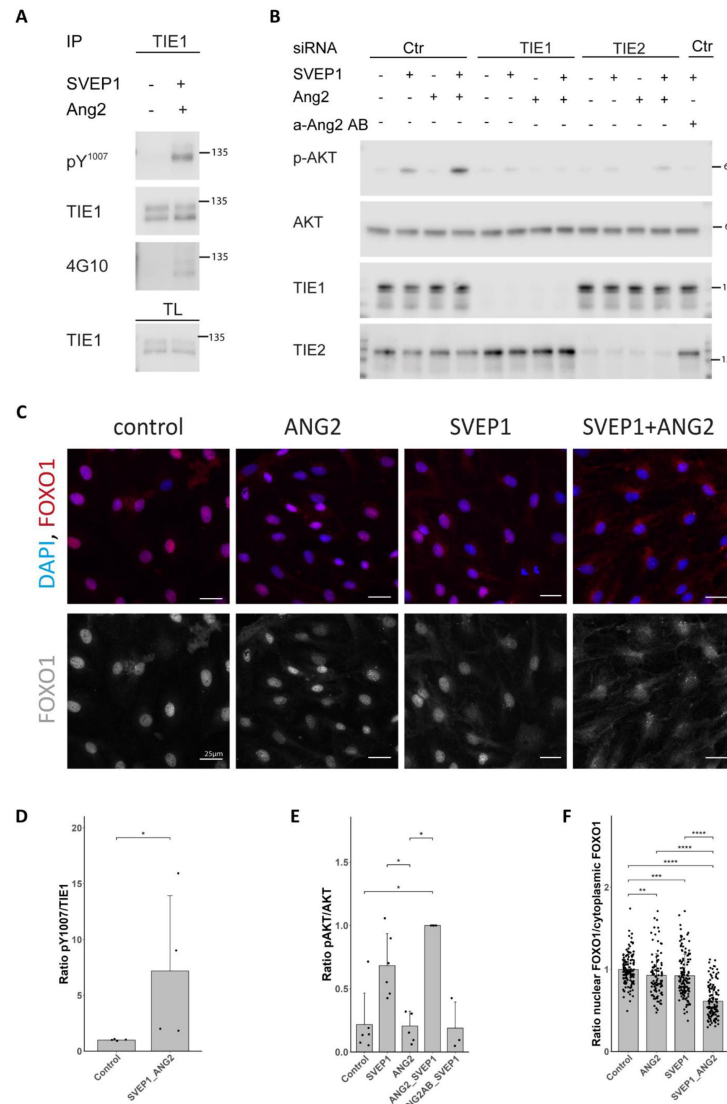


Figure 4

Stimulation of LECs with SVEP1 and ANG2 together leads to phosphorylation of TIE1 and AKT as well as nuclear exclusion of FOXO1 *in vitro*.

(A) Western blot analysis of TIE1 phosphorylation at Y¹⁰⁰⁷ residue in SVEP1 and ANG2 stimulated LECs after immunoprecipitation of TIE1. TIE1 is phosphorylated after stimulation with SVEP1 and ANG2 together. The antibody anti-phospho-TIE2 (pY992, AF2720 R&D Systems) detects both, TIE1 phosphorylation (at Y1007 residue) and TIE2 phosphorylation (at Y992 residue), but can be distinguished by size (Brouillard et al. 2024). (B) Western blot analysis of p-AKT and AKT in the cell lysate of in SVEP1 and ANG2 stimulated LECs. Knock-down of either TIE1 or TIE2 leads to reduced induction of phosphorylation of AKT by SVEP1-ANG2 (C) FOXO1 (red) and DAPI (blue) staining of LECs stimulated with only SVEP1 or ANG2, or in combination. FOXO1 staining is restricted to the nucleus of unstimulated (control) LECs, whereas in cells stimulated with SVEP1 and ANG2 together FOXO1 is in the cytoplasm. (D) Quantification of the pTIE1/total TIE1 ratios shown in A. (N = 4; *p = 0.0211; Mann-Whitney U test; Values are presented as means ± SD) (E) Quantification of the pAKT/total AKT ratios shown in B. Values are presented as means ± SD, N = 3/6; Mann-Whitney U test; Control vs. ANG2_SVEP1 p.adj = 0.028 *; SVEP1 vs. ANG2 p.adj = 0.022 *; ANG2 vs. ANG2_SVEP1 p.adj = 0.028 *; ANG2_SVEP1 vs. aANG2AB_SVEP1 p.adj = 0.091. (F) The ratio of nuclear FOXO1/cytoplasmic FOXO1 decreases after stimulation with either ANG2 or SVEP1. Stimulation of LECs with simultaneous SVEP1 and ANG2 leads to a further reduction of the nuclear FOXO1/cytoplasmic FOXO1 ratio. Each datapoint represents the mean value per image of the ratio of nuclear to cytoplasmic FOXO1 per cell (Mann-Whitney U test; Values are presented as means ± SD; N=4).

known to dimerize via their membrane-proximal D7 Fn3 domains (captured by X-ray structures 5N06 and 5MYB (Leppänen, Saharinen, and Alitalo 2017 [↗](#))), and the 'zippering' together of TIE1 chains in particular, could lead to large oligomeric assemblies with a simultaneous increase in prospective TIE clusters fails to extend the oligomer, because of two reasons: (1) the TIE2 headpiece D1-D2 domains only transiently bind SVEP1's CCP20 module, and fail to capture ANG1 or ANG2, and (2) the 'closed' D4 conformation in TIE2 cannot bind the 'open' TIE1 D4, nor participate in the domain-swapping that cements the TIE dimer in the interaction hub. Whether TIE2 can bind to other parts of SVEP1 in these complexes still needs to be tested.

Discussion

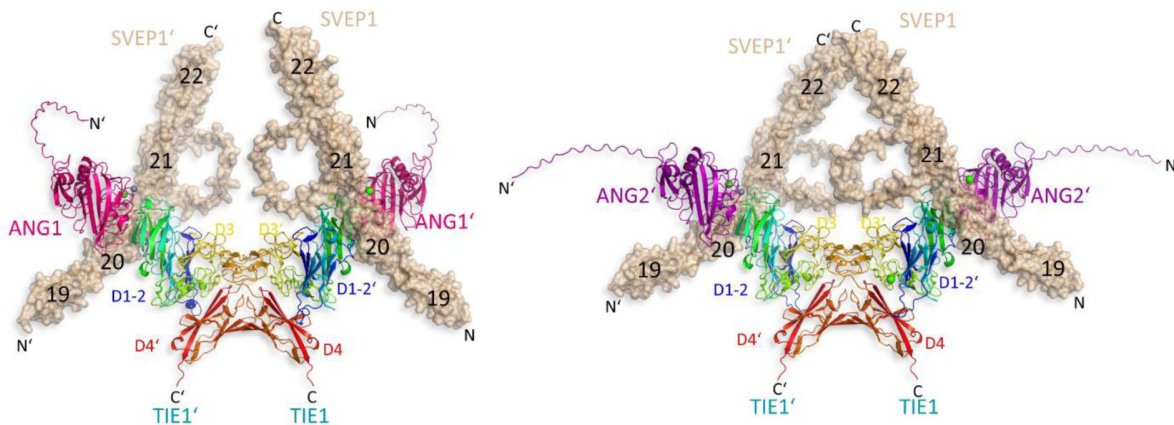
Tie1 has long been considered an orphan receptor. Recently, we and others have shown that *svep1* and *tie1* genetically interact (Hußmann et al. 2023 [↗](#)) and that SVEP1 binds to murine and human TIE1 (Hußmann et al. 2023 [↗](#); Sato-Nishiuchi et al. 2023 [↗](#)). Here we extend the analysis of SVEP1 as a binding ligand for TIE1 with the assistance of AF3 modeling and experimental support that demonstrates potentiated recruitment of TIE1 by SVEP1 in the presence of angiopoietins.

In a binary setup, with no other proteins being present, the CCP19-21 stretch of SVEP1 is a strong binding partner of TIE1, and we were able to quantify the binding strength via Biacore assays. *In vivo* rescue experiments, *in vitro* IP precipitation and modeling data demonstrate that in addition to the CCP20 domain of SVEP1 the CCP5-EGFL7 domain also engages TIE1, and that in the absence of the CCP20 domain (such as in *svep1*^{-/-} zebrafish with a nonsense mutation that leads to a truncated protein) injection of the murine mRNA of the CCP5-EGFL7 domain is sufficient to elicit a biological response. However, the CCP5-EGFL7 domain is not required for the function of *Svep1* during zebrafish lymphatic development, since an in-frame deletion of this fragment does not lead to a phenotype. The binding of CCP5-EGFL7 has a smaller interface compared to the high-affinity binding to CCP20 (Figure 2 [↗](#)) and it is likely that the CCP20 domain, which is the strongest binding site for Tie1, is able to fulfill the function of *Svep1* in a mutant scenario where the CCP5-EGFL7 domain is deleted (*svep1*ΔCCP5-EGFL7 ^{-/-}).

It is known that TIE2 is activated through binding of angiopoietins including ANG1 and ANG2 (Davis et al. 1996 [↗](#); Maisonpierre et al. 1997 [↗](#)) and that in human and mice, angiopoietins do not directly bind to TIE1. In zebrafish however, Ang1 binds to Tie1, Tie2 mutants do not have an obvious cardiovascular phenotype, and the *tie2* locus has been lost in many teleost species (Jiang et al. 2020 [↗](#)). Therefore, zebrafish differ from humans and mice in terms of the TIE pathway, as zebrafish appear to exclusively use Tie1 and not Tie2 (Morooka et al. 2024 [↗](#); Chen, He, and Luo 2022 [↗](#); Jiang et al. 2020 [↗](#)). Both Biacore testing and AF3 modeling revealed that SVEP1 (CCP15-CCP24) does not interact with TIE2. This is in line with transwell migration assays which revealed that SVEP1 induces the migration of lymphatic endothelial cells through binding to TIE1, but not TIE2 (Sato-Nishiuchi et al. 2023 [↗](#)). Whether TIE2 is able to bind SVEP1 in a condition where other proteins like Angiopoietins are present, is still a question of interest and requires further investigation.

We here demonstrate via AF3 model and co-immunoprecipitation, that the affinity of TIE1 for SVEP1 is highly enhanced by the simultaneous binding of the C-terminal domains of ANG2 to the CCP20 module thus generating a complex of SVEP1, Angiopoietin and TIE1 molecules in which the Angiopoietins are able to also bind TIE1. Utilizing AF3 modeling, we were able to confidently predict an interaction hub with a 2:2:2 complex of SVEP1, TIE1 and ANG1/2, where ANG1/2 binds different SVEP1 molecules by cross-lacing of N-terminal coiled-coil dimers which then brings several TIE1 receptors into spatial proximity on the cell surface. This could lead to a cumulative increase in the signal strength of the cytosolic kinase TIE1.

A



B

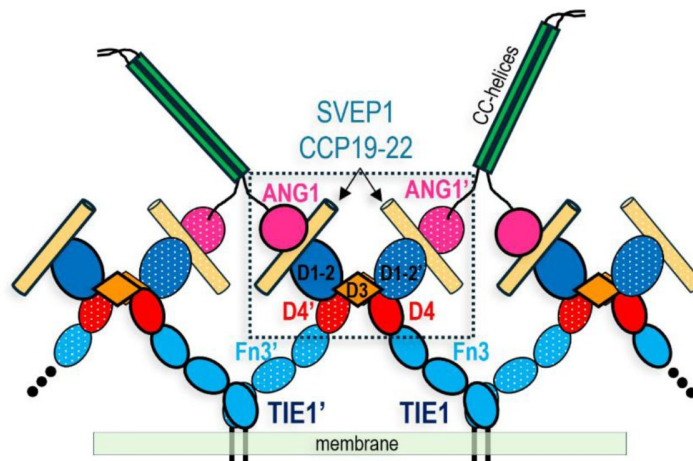


Figure 5

SVEP1, TIE1 and ANG1/2 can form 2:2:2 complexes mediated by SVEP1.

(A) Larger hexameric complexes of SVEP1-TIE1 D1-D4-ANG1/2 in a 2:2:2 stoichiometry could be confidently modeled with AF3. They show a symmetric assembly that is consistent with an interaction hub (in a dotted line box in the adjacent schematic B) of adjacent TIE1 chains that are bound to ANG1 or ANG2 by inclusion of the SVEP1 chain (as shown here, the CCP19-22 stretch). Intriguingly, the TIE1 and TIE1' chains show domain-swapped D4 modules as the chains cross, leading to the following schematic model. (B) Model of SVEP1 enabling the multimerization of TIE1 receptors on the cell surface (with concomitant increase in cytosolic kinase signaling strength) by cross-lacing of N-terminal coiled-coil dimers (or greater assemblies) of ANG1 or ANG2. The linker chain between the C-terminus of the coiled-coil helix and the globular domain is too short to enable the cross-linking of ANG1/2 domains mounted with TIE1 on the angled SVEP1 chains, suggesting that ANG1/2 dimers bridge or glue consecutive TIE1 chains atop the cell surface.

So far, phosphorylation of TIE1 has not been detected when cells were stimulated with SVEP1 or ANG2 only (Saharinen et al. 2005 [\[1\]](#); Sato-Nishiuchi et al. 2023 [\[2\]](#); Savant et al. 2015 [\[3\]](#)). Sato-Nishiuchi et al. (2023) [\[2\]](#) could not detect an increase of the baseline phosphorylation of murine TIE1 in either LECs or 293-F cells that were treated with Svp1. Only TIE2 co-expression increased TIE1 phosphorylation irrespective of the presence or absence of Svp1. In blood and lymphatic endothelial cells, only COMP-Ang1 and ANG1 have been shown to stimulate the phosphorylation of TIE1 (Saharinen et al. 2005 [\[1\]](#); Savant et al. 2015 [\[3\]](#); Yuan et al. 2007 [\[4\]](#)). Co-transfection of TIE1 and TIE2 enhances TIE1 activation and phosphorylation due to heteromeric TIE1-TIE2 complexes. On the contrary, TIE2 phosphorylation is not enhanced by the presence of TIE1 (Saharinen et al. 2005 [\[1\]](#)) and TIE1 is not phosphorylated in the absence of TIE2 upon ANG1 stimulation (Savant et al. 2015 [\[3\]](#)). According to current knowledge, the phosphorylation of TIE1 is therefore dependent on TIE2. In zebrafish however, it was shown that Tie1 can be phosphorylated (Morooka et al. 2024 [\[5\]](#)), and that Tie1, not Tie2, is the relevant receptor for lymphangiogenesis during early development. Furthermore it was reported that autophosphorylation of TIE1 (Y1007) is not dependent on TIE2 but on TIE1, because substitutions in the kinase domain of TIE1 results in a loss of its baseline phosphorylation (Brouillard et al. 2024 [\[6\]](#)). Additionally, after COMP-Ang1 stimulation of TIE1-transfected 293T cells that do not express TIE2, weak phosphorylation is observed, whereby the activation is mediated by ANG1 and not by the COMP domain, a minimal coiled-coil segment that oligomerizes the ANG1 C-term module (Saharinen et al. 2005 [\[1\]](#)), which suggests that the phosphorylation is not dependent on TIE1/TIE2 heterodimerization. Nevertheless, since TIE1 was considered as an orphan receptor during that time, the result could not be explained in 2005 and it was speculated, whether COMP-Ang1-induced Tie1 activation could involve complex formation with additional molecules.

Our results show phosphorylation of TIE1 after combined SVEP1-ANG2 stimulation of hdLECs. SVEP1 proximity to LECs (Karpanen et al. 2017 [\[7\]](#); Hußmann et al. 2023 [\[8\]](#); Wang et al. 2020 [\[9\]](#)). According to data from the human protein atlas (<https://www.proteinatlas.org> [\[10\]](#)), SVEP1 is also not expressed in HEK293 cells. Since SVEP1, which is necessary for the phosphorylation of TIE1, is provided by cells in the immediate vicinity of ECs and not by ECs themselves, the phosphorylation of TIE1 after ANG2 stimulation was never detected in cell culture experiments without SVEP1 being present. Whether TIE2 is necessary for the phosphorylation of TIE1 after SVEP1-ANG2 treatment needs further investigation. Our model shows that after binding of SVEP1 to TIE1, ANG1/2 are then able to dock onto this complex, and serve as a molecular glue for both SVEP1 and TIE1. Thus, TIE1-bound SVEP1 would serve as a cell-surface-tethered interaction hub for angiopoietins that are oligomerized – by either the native coiled-coil extensions, or the minimal COMP domain – and create cross-linked fields of signaling TIE1 receptors (**Figure 5** [\[11\]](#)).

The PI3K/Akt signaling pathway is involved in Svp1 induced LEC migration, as LECs show an increased Akt phosphorylation after Svp1 treatment (Sato-Nishiuchi et al. 2023 [\[2\]](#)). We were able to show that combined SVEP1-ANG2 stimulation leads to an even higher phosphorylation of AKT in human LECs than the sum of SVEP1 or ANG2 stimulations alone. Blocking of ANG2 reduces AKT phosphorylation after stimulation with SVEP1, signifying that SVEP1 cannot induce downstream signaling on its own, but only in combination with angiopoietins.

Furthermore, both TIE1 and TIE2 are required for the full signaling potential of SVEP1-ANG2, as knock-down of either TIE1 or TIE2 did reduce the phosphorylation of AKT upon SVEP1-ANG2 stimulation. A signaling event downstream of PI3K/Akt is the inactivation and nuclear exclusion of FOXO1 (Daly et al. 2004 [\[12\]](#)), which is reduced in Tie1-silenced vascular endothelial cells (Korhonen et al. 2016 [\[13\]](#)). Furthermore, SVEP1 induces FOXO1 nuclear exclusion. This was shown in LECs as well as *in situ* where FOXO1 immunostaining of lymphatic vessels in wild type mice was detected in the nucleus, whereas in *Svp1* deficient mice, FOXO1 staining was detectable in both the nucleus and cytoplasm (Sato-Nishiuchi et al. 2023 [\[2\]](#)). During secondary sprouting of endothelial cells in the zebrafish trunk, Tie1 induces nuclear exclusion of *foxo1a* since *tie1* mutants displayed an enhanced localization of *foxo1a* in the nucleus (Morooka et al. 2024 [\[5\]](#)). Thus, nuclear versus

cytoplasmic ratio of FOXO1 is used as a readout for signaling in a number of systems. Here we show that stimulation of LECs with a combination of SVEP1 and ANG2 leads to a stronger nuclear exclusion of FOXO1 than with either factor alone, thus indicating that stronger binding of SVEP1 to TIE1 also potentiates downstream signaling of TIE1.

SVEP1's CCP20 module acts as a critical binding partner for TIE1, but not TIE2, when interacting with ANG1/2. However, it remains unclear whether TIE2 binds to other sites of SVEP1 since in the Biacore and the immunoprecipitation assays the C-terminus of the SVEP1 protein, starting from CCP15, was both are necessary for SVEP1 stimulated phosphorylation of AKT. Furthermore, in capillaries, only depletion of *Tie1*, but not of *Tie2* had an effect, whereas in collecting vessels, depletion of *Tie2* resulted in a phenotype (Korhonen et al. 2022 [DOI](#)). This result can most likely be explained by high expression levels of TIE2 in collecting vessels, but low expression levels of TIE2 in capillaries. Additionally, it is proposed that SVEP1 acts as a modifier of TIE2 expression during Schlemm's canal development as patients that are trans-heterozygous for TIE2 and SVEP1 develop glaucoma and the p.R997C-SVEP1 variant, which is mutated in some glaucoma patients and located shortly after the furin cleavage site, fails to enhance TIE2 expression in HEK293T cells compared to the wild type SVEP1 (Young et al. 2020 [DOI](#)). Depending on the downstream signaling event as well as the lymphatic cell type (capillaries versus collecting vessels), TIE2 might be needed for phosphorylation of TIE1 either independently of SVEP1 or binding to other domains of SVEP1 than CCP19-21.

The activation mechanisms of the TIE receptors have long been a point of contention. Especially TIE1 activation and how possible signaling occurs has remained enigmatic, and only recently SVEP1 has been demonstrated to serve as a ligand for the TIE1 transmembrane receptor, which for many years had been considered an orphan receptor with phosphorylation believed to be dependent on TIE2. We here show that SVEP1-ANG2 is able to cause phosphorylation of TIE1 and show that the presence of ANG2 results in stronger recruitment of TIE1 to SVEP1, resulting in a model that predicts a higher order complex of SVEP1, TIE1 and ANG1/2. In such complexes, ANG1/2 binds different SVEP1 molecules by cross-lacing of N-terminal coiled-coil dimers which then brings several TIE1 receptors into spatial proximity on the cell surface. This could lead to a signaling hub, where the signal strength of the cytosolic kinase TIE1, or another signaling moiety, is increased.

Material and Methods

Key Resources Table

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
genetic reagent (<i>D. rerio</i>)	<i>Tg(flt4:mCitrine)^{hu}₇₁₃₅</i>	van Impel et al., 2014	ZFIN: hu7135	
genetic reagent (<i>D. rerio</i>)	<i>svep1^{hu6123}</i>	Karpanen et al., 2017	ZFIN: hu6123	
genetic reagent (<i>D. rerio</i>)	<i>svep1^{hu4767}</i>	Karpanen et al., 2017	ZFIN: hu4767	
genetic reagent (<i>D. rerio</i>)	<i>svep1ΔCCP5-EGF7^{mu424}</i>	This paper	ZFIN: mu424	
cell line (<i>homo sapiens</i>)	HEK293 EBNA	Manuel Koch		
cell line (<i>homo sapiens</i>)	293T HEK cells	Roukens et al., 2015		
cell line (<i>homo sapiens</i>)	Human Dermal Lymphatic Endothelial Cells	PromoCell	C-12216	
transfected construct (<i>homo sapiens</i>)	TIE1-HA in PCEP4	Hußmann et al., 2023		TIE1 cDNA provided by Hellmut Augustin
transfected construct (<i>homo sapiens</i>)	TIE2-HA in PCEP4	Hußmann et al., 2023		provided by Manuel Koch
transfected construct (<i>homo sapiens</i>)	ANG1-FLAG in	This paper		provided by Manuel Koch
transfected construct	ANG2-FLAG in	This paper		provided by

(<i>homo sapiens</i>)				Manuel Koch
Peptide, recombinant protein	VEGF-C	R&D	9199-VC	
Peptide, recombinant protein	Human recombinant protein ANG1	R&D Systems	923-AN-025	
Peptide, recombinant protein	Human recombinant protein ANG2	R&D Systems	623-AN-025	
Peptide, recombinant protein	SVEP1-Strep II 150kDa Purified recombinant protein	Hußmann et al., 2023		provided by Manuel Koch
Peptide, recombinant protein	SVEP1-Strep II domain 20_21 Purified recombinant protein	This paper		provided by Manuel Koch
Peptide, recombinant protein	SVEP1-Strep II domain 19_20_21 Purified recombinant protein	This paper		provided by Manuel Koch
Peptide, recombinant protein	SVEP1-Strep II domain 19_20 Purified recombinant protein	This paper		provided by Manuel Koch
Peptide, recombinant protein	SVEP1-Strep II domain 5/6/7 Purified recombinant protein	This paper		provided by Manuel Koch
Peptide, recombinant protein	SVEP1-Strep II domain 19_20mut_21 Purified recombinant protein	This paper		provided by Manuel Koch

Peptide, recombinant protein	VEGFC-Strep II Purified recombinant protein	This Paper		provided by Manuel Koch
antibody	anti-HA (rat, monoclonal)	Roche	11867423001	1:10000
antibody	anti-FLAG (rabbit, polyclonal)	Sigma-Aldrich	F1804	1:10000
antibody	Strep-Tactin® HRP conjugate	Iba-lifesciences	2-1502-001	1:10000
antibody	anti-goat (donkey, polyclonal) 488	Thermo Scientific	A11055	1:500
antibody	anti-mouse (donkey, polyclonal) 546	Thermo Scientific	A-21123	1:500
antibody	anti-rabbit (donkey, polyclonal) 647	Thermo Scientific	A31573	1:500
antibody	anti-mouse (donkey, polyclonal) HRP	Jackson Immuno Research	715-035-150	1:15000
antibody	anti-goat (donkey, polyclonal) HRP	Immuno Reagents Inc.	DkxGt-003-EHRPX	1:15000
antibody	anti-rabbit (donkey polyclonal) HRP	Jackson Immuno Research	711-035-152	1:15000
antibody	anti-TIE2 (goat, polyclonal)	R&D Systems	AF313	1:300
antibody	anti-TIE1 (goat, polyclonal)	R&D Systems	AF619	1:300
antibody	anti-phospho-TIE2 (rabbit, polyclonal)	R&D Systems	pY992, AF2720	1:300

antibody	anti-phosphotyrosine (mouse, monoclonal)	Millipore	05-321, clone 4G10,	1:500
antibody	Phospho-AKT (S473) (rabbit, polyclonal)	Cell Signaling Technology	9271	
antibody	anti-total AKT (mouse, monoclonal) anti-total AKT (rabbit, polyclonal)	Cell Signaling Technology	2920, 9272	
antibody	phospho-ERK	Cell Signaling Technology	43705	1:2000
antibody	total ERK (mouse, monoclonal)	Cell Signaling Technology	4696	1:4000
antibody	ZO-1 (mouse, monoclonal)	Thermo Scientific	33-9100	1:300
antibody	FOXO1(rabbit, monoclonal)	Cell Signaling	C29H4, 2880	1:300
antibody	HSC70 (mouse, monoclonal)	Sigma-Aldrich	MABE11 20	1:1000
antibody	Anti-ANGPT2 (MEDI3617)	Selleckchem	A2770	
chemical compound, drug	Pierce™Avidin	Thermo Scientific	21121	
chemical compound, drug	Endothelial Cell Growth Medium MV2	PromoCell	C-22022	
chemical compound, drug	Endothelial Cell Growth Medium MV2	PromoCell	C-39226	

chemical compound, drug	FCS	Merck Chemicals GmbH	F7524	
chemical compound, drug	DMEM/F-12, GlutaMAX™ Supplement	ThermoFisher Scientific	10565018	
chemical compound, drug	fibronectin (1ug/mL)	Sigma	F1141-1MG	
chemical compound, drug	SuperSignal™ West Pico PLUS Chemilumineszenz-Substrat	Thermo Fisher Scientific	34580	
chemical compound, drug	Fluormount-G	Thermo scientific	00-4958-02	
chemical compound, drug	Strep-Tactin® Superflow® high capacity resin	IBA Lifesciences GmbH	2-1208-002	
chemical compound, drug	Lipofectamin 2000 transfection reagent	ThermoFisher Scientific	11668030	
chemical compound, drug	1-phenyl-2-thiourea (75 mM)	Sigma	P7629	
chemical compound, drug	tricaine	Sigma	A5040	
chemical compound, drug	0.8% low melting agarose	Thermo Fischer	16520100	
chemical compound, drug	protein G–Sepharese	GE Healthsciences	GE17-0618-01	
chemical compound, drug	PrimeSTAR® Max DNA Polymerase	TaKaRa	R045A	
Commercial assay, kit	QIAquick PCR purification kit	QIAGEN	28106	

Commercial assay, kit	GeneJET gel extraction kit	ThermoScientific	K0503	
chemical compound, drug	DpnI,	Promega	R623A	
chemical compound, drug	AvrII	New England Biolabs	R0174S	
chemical compound, drug	T4 DNA Ligase (1-3u/μl)	Promega	M180A	
chemical compound, drug	T7 RiboMAX TM Large Scale RNA Production System	Promega	#P1280	
chemical compound, drug	Ribo m7G Cap Analog	Promega	#P1711	
chemical compound, drug	2x rapid ligation buffer	Promega	#C671A	
chemical compound, drug	10x ligation buffer	Promega	#C126B	
cells	Promega JM109 Competent E. coli	Promega	#L1001	
cells	NEB® 5-alpha Competent E. coli	Promega	#C2987	
Other	wtSvep1	Sato-Nishiuchi et al. 2012		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_PoIC	Sato-Nishiuchi et al. 2012		Construct for mRNA injection (<i>Mus musculus</i>)

Other	Svep1_dN-EGFL6	Sato-Nishiuchi et al. 2012		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_dPTX	Sato-Nishiuchi et al. 2012		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_dN-CCP21	Sato-Nishiuchi et al. 2012		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_dPTXdCCP21	This paper		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_dPTXd22-C	This paper		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_dPTXd7-34	This paper		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_dN-EGF7	This paper		Construct for mRNA injection (<i>Mus musculus</i>)
Other	Svep1_CCP5-EGF7	This paper		Construct for mRNA injection (<i>Mus musculus</i>)
software, algorithm	R Studio	2023.06.0 Build 421		

software, algorithm	Fiji-ImageJ	(version 1.54 f)		
software, algorithm	ZEISS Elyra 7 superresolution microscope			
software, algorithm	ZEN Microscopy Software	Zen blue and Zen black		
software, algorithm	Leica	LAS X 3.5.7.23225		
software, algorithm	Adobe Illustrator	2023		
software, algorithm	Arivis	Vision 4D 4.1		
software, algorithm	AlphaFold2-multimer (AF2.3.2)	accessed via ColabFold 1.5.5 (https://github.com/sokr ypton/ColabFold)		
software, algorithm	DeepMind portal	(https://alphafoldserver.com/).		
software, algorithm	Predictome web site	https://predictomes.org/tools/af3/		
software, algorithm	PyMOL3.04	http://www.pymol.org		
Sequence-based reagent	svep1_in567del_fwd	This paper	PCR primer: Genotyping svep1CC P5-EGF7 ^{mu42} ₄	AAGAACC AGAGAG AGGCACT G
Sequence-based reagent	svep1_EGF1_rv	This paper	PCR primer: Genotyping svep1CC	AAAGGAA TGGCAGA CTGGTC

			P5-EGF7 ^{mu42} ₄	
Sequence-based reagent	svep1_CCP5_fw1	This paper	PCR primer: Genotyping <i>svep1</i> CC P5-EGF7 ^{mu42} ₄	GCCCTAA TATATAT GTAGAAC TCCT
Sequence-based reagent	gRNA upstream: CCP 5 1.1	This paper	CRISPR/Cas9 <i>svep1</i> CC P5-EGF7 ^{mu42} ₄	TCCAGTA AAGCTGT GAGCCC
Sequence-based reagent	gRNA downstream: EGF 1.1	This paper	CRISPR/Cas9 <i>svep1</i> CC P5-EGF7 ^{mu42} ₄	GGGGAT GGAAGA AACTGCA C
Sequence-based reagent	Δ21_fw	This paper	PCR primer: ΔN-PTXΔCCP21	ATTTCCTA TTGAATG TGATTTG C
Sequence-based reagent	Δ21_rev	This paper	PCR primer: ΔN-PTXΔCCP21	ATGTTGA GGGTGA GCCAGAT A
Sequence-based reagent	Δ22-end_fw	This paper	PCR primer: ΔN-PTXΔCCP22-C	CCACCAT CACCACC ATTAG
Sequence-based reagent	Δ22-end_rev	This paper	ΔN-PTXΔCCP22-C	TCAATGG AAATGCA AGAGG
Sequence-based reagent	Δ7-34_fw	This paper	PCR primer: ΔN-	GTCTGTC GGTTCCC ATGTC

			PTXΔCCP 7-34	
Sequence- based reagent	Δ7-34_rev	This paper	PCR primer: ΔN- PTXΔCCP 7-34	TGCACAG TTTTTCCC ATCTC
Sequence- based reagent	Δ567_fw	This paper	PCR primer: ΔN-EGF7	GAACCTG TAAATG TAAGGCT CCA
Sequence- based reagent	Δ567_rv	This paper	PCR primer: ΔN-EGF7	ATCAGAG CAGAACA TGCTGCA GCT
Sequence- based reagent	Δ22-end_fw	This paper	PCR primer: CCP5- EGF7	CCACCAT CACCACC ATTAG
Sequence- based reagent	Δ7-34_rev	This paper	PCR primer: CCP5- EGF7	TGCACAG TTTTTCCC ATCTC
Sequence- based reagent	svep1hu6123_fwd 1	This paper	KASPR primer: Genotypi ng <i>svep1</i> ^{hu61 23}	GAAGGT GACCAAG TTCATGC TCCTGGT AGCACAG ATATATC AGACTGT
Sequence- based reagent	svep1hu6123_fwd 2	This paper	KASPR primer: Genotypi ng <i>svep1</i> ^{hu61 23}	GAAGGTC GGAGTCA ACGGATT CCTGGTA GCACAGA TATATCA GACTGA
Sequence- based reagent	svep1hu6123_rev	This paper	KASPR primer: Genotypi ng <i>svep1</i> ^{hu61 23}	GTTGTTG TGTGGTC TGTAGCC TTCC

Sequence-based reagent	svep1hu4767_fwd 1	This paper	KASPR primer: Genotyping <i>svep1</i> ^{hu4767}	GAAGGT GACCAAG TTCATGC TGGGGA GATGATG TCTCTTG CA
Sequence-based reagent	svep1hu4767_fwd 2	This paper	KASPR primer: Genotyping <i>svep1</i> ^{hu4767}	GAAGGTC GGAGTCA ACGGATT GGGGAG ATGATGT CTCTTGC T
	svep1hu4767_rev	This paper	KASPR primer: Genotyping <i>svep1</i> ^{hu4767}	CAGGCAC TGTGCAG GTAAAGT CATT
Sequence-based reagent	TIE1 siRNA	IDT	234070784	hs.Ri.TIE1.13
Sequence-based reagent	TEK siRNA	Thermo Fisher Scientific	Catalog #4390824, Catalog #4392420,	siRNA ID #s13984, siRNA ID #s224717
Other	Immobilon-P Membrane, PVDF	Millipore Thermo	IPVH00010	

Zebrafish strains

Research on animals was conducted according to the guidelines of the animal ethics committees of the University of Münster, Germany. Zebrafish strains were maintained under standard husbandry conditions according to FELASA guidelines (Aleström et al. 2020). In this study, the following transgenic and mutant lines have been used: *svep1*^{hu4767} (Karpanen et al. 2017), *svep1*^{hu6123} (Karpanen et al. 2017), *Tg(flt4BAC:mCitrine)*^{hu7135} (van Impel et al. 2014).

In frame deletion of *svep1*ΔCCP5-EGF7^{mu424} (-1049 +38 bp, 6995bp after the first ATG, ENSDARG00000013526) was induced using CRISPR/Cas9 technology with following sgRNAs:

gRNA upstream: CCP 5 1.1 TCCAGTAAAGCTGTGAGCCC

gRNA downstream: EGF 1.1 GGGGATGGAAGAACTGCAC

Genotyping

For genotyping of *svep1*^{hu6123} and *svep1*^{hu4767} KASPar (Biosearch Technologies) was used (Primers in Key Resources Table). The *svep1*CCP5-EGF7^{mu424} mutants were genotyped using a three primer PCR (Suppl. Table 1) with a standard PCR protocol.

Generation of murine deletion constructs of *Svep1*

Following cDNA constructs were kindly provided by Sato-Nishiuchi: wt*Svep1*, PolC, dN-EGFL6, dPTX and dN-CCP21 (Sato-Nishiuchi et al. 2012 [DOI](#)).

Cloning

The deletion constructs were amplified with primers (Key Resources Table), which were located on the left and right site of the to-be-deleted region and were orientated away from this region. Therefore, everything except the to-be-deleted region was amplified. For long-range PCR amplification the PrimeSTAR® Max DNA Polymerase (TaKaRa, #R045A) and 20 ng of the template plasmid was used. For dPTXdCCP21, dPTXd22-C, dPTXd7-34 and dN-EGF7, dPTX was used as a template. For CCP5-EGF7 the dPTXd7-34 construct was used as a template. 2 µl of the PCR product was run on a 1% agarose gel stained with ethidiumbromide to test if there were side products which were also amplified during the PCR. If there was only one band and inferentially no side products, the PCR product was purified by the QIAquick PCR purification kit (QIAGEN, #28106). If there were more bands, the PCR product was purified with the GeneJET gel extraction kit (ThermoScientific, #K0503) after running the PCR product on a 1% gel and cutting out the band with the expected size. Before the PCR purification of the deletion constructs with the QIAquick kit, the PCR product was digested with 2 µl DpnI (Promega, #R623A) for more than 4 hours to get rid of the original plasmid. According to the primers for the deletion constructs, (see Key Resources Table) the ends of the PCR product were blunt ended and phosphorylated at the 5' end. The purified PCR product (50 ng) was ligated with the T4 DNA Ligase (1-3 u/µl) (Promega, #M180A) for 4 hours at room temperature with 2x rapid ligation buffer (Promega, #C671A), or overnight at 4°C using 10x ligation buffer (Promega, #C126B). The ligated constructs were transformed in either Promega JM109 Competent *E. coli* (Promega, #L1001) or NEB® 5-alpha Competent *E. coli* (Promega, #C2987) for higher efficiency and NEBuilder reactions. Using colony PCR, positive clones were selected and further amplified.

mRNA generation and injection

To generate mRNA for injections, 10 µg of the plasmid DNA was digested with AvrII (New England Biolabs, #R0174S) and purified with the QIAquick PCR purification kit. The mRNA was generated with 2 µg of the linearized vector and the T7 RiboMAX™ Large Scale RNA Production System (Promega, #P1280). To protect the mRNA from intracellular degradation and to enhance translation efficiency, Ribo m7G Cap Analog (Promega, #P1711) was added. Embryos from a *svep1* heterozygous in-cross were injected in the yolk immediately after egg deposition, using 1 ng of the mRNA per embryo. For the rescue experiments, mRNA was always injected in eggs from single pairs with comparable phenotypic strength. For ΔPTXΔ7-34 we could not get enough non-injected controls and therefore compared the injected embryos to non-injected embryos derived from the same parents of another experiment. The rescue efficiency was determined by counting fragments of TD at 5 dpf. The non-injected mutants normally display 0-4 fragments of TD.

Microscopy

Live imaging was carried out on 5 dpf embryos. Before 24 hpf, 1-phenyl-2-thiourea (75 mM, Sigma, #P7629) was added to inhibit melanogenesis. For imaging, embryos were anesthetized with 0.0168% tricaine (Sigma, #A5040) and embedded in 0.8% low melting agarose (Thermo Fischer, #16520100) dissolved in embryo medium. Embryos were imaged with an inverted Leica SP8 microscope using a ×20/×0.75 or ×10/×0.75 dry objective detection and employing Leica LAS X 3.5.7.23225 software. Confocal stacks were processed using Fiji-ImageJ version 1.54 f. Images and figures were assembled using Adobe Illustrator. All data were processed using raw images with brightness, color, and contrast adjusted for printing. Scoring of TD fragments was performed using a Leica M165 FC and an X-Cite 200DC (Lumen Dynamics) fluorescent light source.

Recombinant protein expression

SVEP1 active (GenBank: NM_153366, aa: 2261-2890), SVEP1 C-term (GenBank: NM_153366, aa: 2261-3571), ectodomain Tie1 (GenBank: NM_005424, aa: 21-760), and ectodomain Tie2 (GenBank: NM_000459, aa: 23-745) were amplified from human cDNA by PCR. PCR products were cloned into a modified sleeping beauty transposon expression vector containing a BM40 signal peptide sequence and a Twin-Strep-tag. For recombinant protein production, stable HEK293 EBNA cell lines were generated employing the sleeping beauty transposon system (Kowarz et al., 2015). Briefly, expression constructs were co-transfected with a transposase plasmid (1:10) into the HEK293 EBNA cells using FuGENE HD transfection reagent (Promega). After selection with puromycin (2 µg/ml), cells were induced with doxycycline. Supernatants were filtered and the recombinant proteins purified via Strep-Tactin®XT (IBA Lifescience) resin. Proteins were then eluted by biotin-containing TBS-buffer (IBA Lifescience). The human sequences of angiopoietin-1 (NM_001199859, aa: 20 – 497) and angiopoietin-2 (NM_001118887, aa: 20–495) were cloned into the PCEP episomal expression system (transient) including an N-terminal Flag-tag sequence.

Surface Plasmon Resonance

Protein-protein interactions were analyzed by surface plasmon resonance (SPR) experiments using a Biacore T200 system (Cytiva, Uppsala, Sweden). For immobilization of hSVEP1, a CM5 sensor chip was activated with N-hydroxysuccinimide (NHS) and 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC) according to the manufacturer's instructions, and 2050 response units (RU) of SVEP1 active were coupled in 10 mM sodium acetate buffer, pH 4.0, at a flow rate of 10 µL/min. For the interaction measurements, the analytes, Tie1 and Tie2, were passed over the sensor chip at a flow rate of 30 µL/min in running buffer (10 mM HEPES, 150 mM NaCl and 0.005% Tween20). The binding kinetics were calculated by fitting a 1:1 interaction model using the BIAevaluation software (Cytiva, Uppsala, Sweden).

Negative staining

The interaction of SVEP1 C-term with angiopoietin was visualized by negative staining and transmission electron microscopy. Briefly, the proteins were incubated for 1 hour at 37°C in Tris-buffered saline (TBS), pH 7.4. Prior the visualization in the electron microscope, the proteins were negatively stained with uranyl formate on the grids. Specimens were examined in a Philips/FEI CM 100 TWIN transmission electron microscope operated at 60 kV accelerating voltage. Images were recorded with a side-mounted Olympus Veleta camera with a resolution of 2048 × 2048 pixels (2k × 2K) and the ITEM acquisitions software.

Cell culture and treatment/stimulation

HEK293T cells were cultured in DMEM (Thermo Fisher Scientific #61965026) and seeded one day before transfection utilizing Lipofectamine™ 2000 (Thermo Fisher Scientific, #11668027). hDLECs (Human Dermal Lymphatic Endothelial Cells, PromoCell) were cultured in Endothelial Cell Growth Medium MV2 (PromoCell, #C-22022) on fibronectin (Sigma, #F1141-1MG, 1µg/mL)-coated culture flasks. LEC culture medium was supplemented with 25 ng/ml VEGF-C (R&D, #9199-VC). Cells were seeded on ibidi slides (ibidi, #81201) and grown to confluency before the treatment.

For knock-down experiments, hDLECs were transfected with Tie1, Tie2 or control siRNA (see key resources table) using Lipofectamine™ RNAiMAX Transfection Reagent one day after splitting and two days prior to analysis of AKT phosphorylation.

For the pull-down assay, different versions of human StrepII tagged SVEP1 protein (amino acids 2261–3571) and a StrepII tagged control protein were incubated with 25 µl Strep-TactinXT 4Flow highcapacity resin (iba-lifesciences, #2-5030-002) in 500 µl binding buffer (50 mM Tris-HCl at pH 7.5, 100 mM NaCl, 0.02% Triton X-100) for 30 min. Subsequently, the cell lysate of TIE1-HA

transfected HEK293T cells was added. After 2 hours of incubation, the beads were washed 5 times with Ripa buffer (50 mM Tris (pH 7.5), 1% NP-40, 0.1% SDS, 0.5% Na-deoxycholate, 150 mM NaCl) and boiled for 5 min at 95°C in sample buffer. For pull down assays that include Angiopoietins, TIE1-HA and ANG2-FLAG/ANG1-FLAG were co-transfected and the cell lysates were pre-cleared with Strep-TactinXT 4Flow high capacity resin, before adding them to the StrepII tagged SVEP1 protein.

For TIE1 autophosphorylation, hdLECs were starved for 3 hours in 0.5% BSA-containing MV2 medium, followed by 1 hour of stimulation with ANG2 and SVEP1 alone or in combination. The cells were lysed in PLCLB lysis buffer (150 mM NaCl, 5% glycerol, 1% Triton X-100, 1.5 mM MgCl₂, 2mM CaCl₂, 50 mM HEPES, pH 7.5, 1 x Complete-EDTA-free proteinase inhibitors) followed by TIE1 immunoprecipitation using anti-TIE1 antibodies (AF619, R&D Systems) and protein G–Sephareose (GE Healthsciences Ab, #GE17-0618-01) at +4°C. After 2 hours of incubation, the beads were washed 5 times with PLCLB lysis buffer and boiled for 5 min at 95°C in sample buffer.

For analysis of pAKT, growth medium of confluent hdLECs was changed to MV2 medium containing growth supplements but lacking VEGFC. Cells were starved for 3 hours in 0.5% BSA-containing MV2 medium, followed by 40-60 minutes of stimulation with ANG2 (0.5 µg/mL) and/or SVEP1 (1 µg/mL). Cells were treated with anti-ANG2 antibody (MEDI3617 A2770, Selleckchem) for 2 h prior to stimulation with SVEP1. The cells were lysed in lysis buffer (20 mM Tris-HCl (pH 7.4), 150 mM NaCl, 2mM CaCl₂, 1.5 mM MgCl₂, 1 mM Na₃VO₄, 1 % (v/v) Triton-X-100, 0.04 % (w/v) NaN₃, 1 x Complete-EDTA-free proteinase inhibitors) or PLCLB lysis buffer and centrifuged before gel analysis.

Western blot

Cell lysates and cell precipitates were subjected to Western blot analysis by separation in SDS-page and blotting onto PVDF membrane. Blots were probed with primary antibodies followed by HRP coupled secondary antibodies and ECL detection (SuperSignal™ West Pico PLUS Chemilumineszenz-Substrat, Thermo Fisher Scientific, 34580). Blots were imaged using a Odyssey FC (LICOR). Primary and secondary antibodies used for analysis are listed in the key resources table and are stated in the figure of the respective blot. Before using the Strep-Tactin® HRP antibody, the membrane was incubated in 2,5µg/ml Avidin (biotin blocking solution) for 10 minutes. HSC70 was used to assure equal loading of the gels and values were normalized by the level of HSC70.

Immunocytochemistry and imaging

Before the treatment, hdLECs were starved for 4 hours in MV2 medium without supplements and with 0,5 % FBS. After treatment with SVEP1 and/or ANG2 (623-AN R&D Systems) for 1-hour cells were fixed PBS for 5 minutes and blocked with 2% BSA for 1 hour. Incubation of the slides with the antibody was conducted over night at 4°C (ZO-1 (Thermo Scientific, 33-9100), FOXO1 (Cell Signaling, C29H4, #2880)). After washing the slides several times with 0,1%PBS-TritonX100 the secondary antibody staining was conducted at room temperature for 50 minutes (Donkey-anti-rabbit-647 (Thermo Scientific A31573), Donkey-anti-mouse-546 (Thermo Scientific), DAPI). Slides were washed and mounted using Fluormount-G (Thermo scientific 00-4958-02). Images were taken using ZEISS Elyra 7 superresolution microscope in Apotome mode with 20x objective and processed with ZEN Microscopy Software (Zen blue 3.5 and Zen black). ZEISS arivis Software was used to calculate the pixel intensities inside the nucleus and in the cytoplasm. ZO1 was used to identify single cells and the ratio was calculated for each cell.

Statistics and reproducibility

Data sets were tested for normality (Shapiro–Wilk) and equal variance. P-values of data sets with normal distribution were determined by Student's *t*-test. In case data values did not show normal distribution, a Mann–Whitney U test was performed instead. For multiple comparison the p values were adjusted with Bonferroni correction. Results are presented as mean ± SD. Statistical tests were performed using GraphPad Prism 8 and R. For visualizations, the R package ggplot2 was used. All experiments were carried out at least two times. Not all *in vivo* rescue experiments were carried out two times, due to limited numbers of embryos. However, injections of ΔN-PTX, CCP5-EGF7, and ΔN-EGF7 were carried out multiple times. P values > 0.05 were considered not significant. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$.

Structure prediction and modeling of SVEP1 complexes

Amino acid sequences for human SVEP1 (<https://www.uniprot.org/uniprotkb/Q4LDE5>), TIE1 and 2 (<https://www.uniprot.org/uniprotkb/P35590> and [/Q02763](https://www.uniprot.org/uniprotkb/Q02763)) and ANG1/2 (<https://www.uniprot.org/uniprotkb/Q15389> and [/O15123](https://www.uniprot.org/uniprotkb/O15123)) were retrieved from UniProtKB. The most recent version of AlphaFold2-multimer (AF2.3.2) was accessed via ColabFold 1.5.5 (<https://github.com/sokrypton/ColabFold>) while the DeepMind portal was used for AF3 analysis (<https://alphafoldserver.com/>). For the template-free AF2.3 predictions, num recycles=20 and num models=10 were used, top ranked models were relaxed with AMBER, and models ranked based on iPTM scores; in turn, default parameters were used for AF3, and PAE plots were visualized using the Predictome web site (<https://predictomes.org/tools/af3/>). Initially, for all proteins, full-length sequences were used as the input for the prediction. Models were analyzed and visualized with PyMOL3.04 (<http://www.pymol.org>).

Data availability

Scripts used for data analysis available at GitHub at https://github.com/MuensterImagingNetwork/Uphoff_et_al_2025/tree/main (copy archived at Münster Imaging Network, 2025). The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

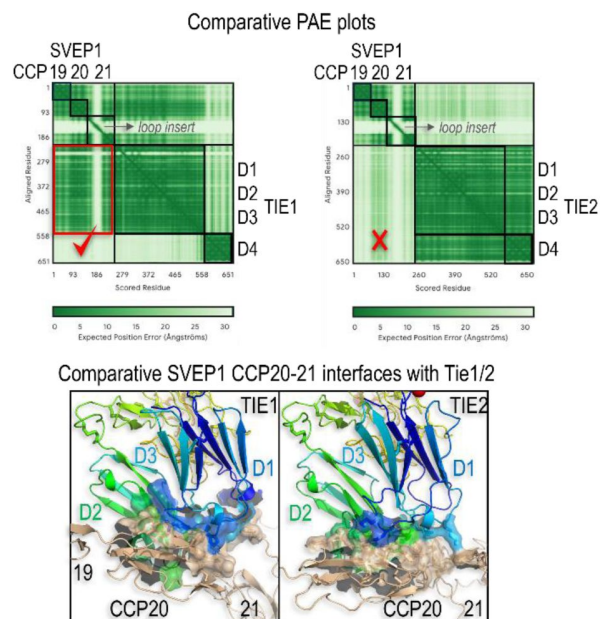
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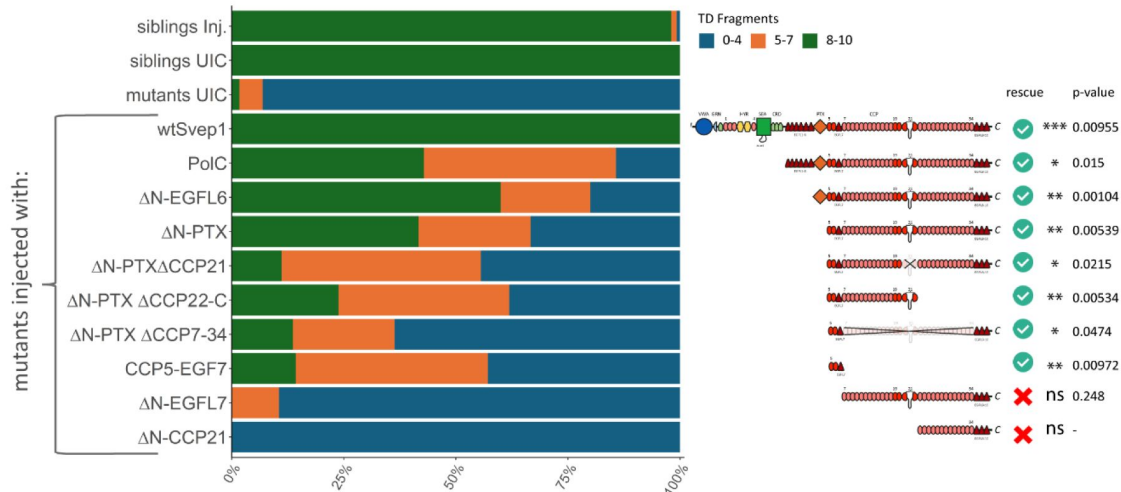


Suppl. Figure 1

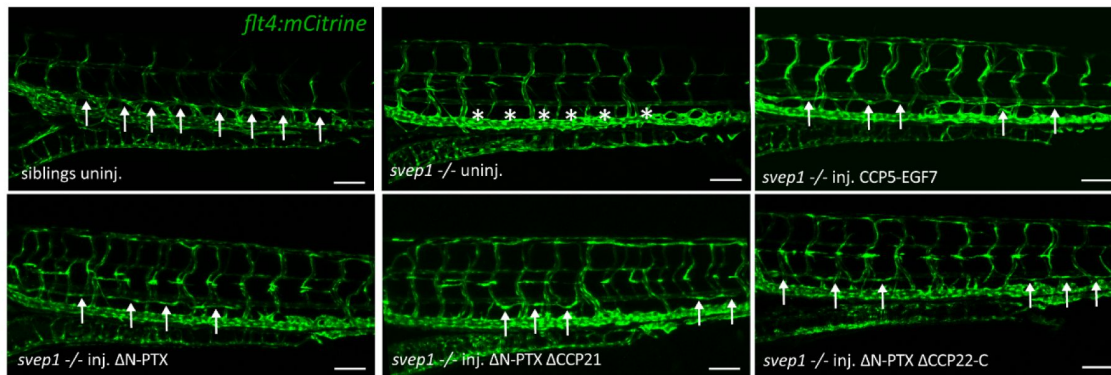
The contrasting interactions of TIE1 and TIE2 D1-D2 domain segments against the CCP19-21 SVEP1 chain are shown by PAE plots that are unequivocally strong for TIE1, and much weaker for TIE2.

This is also seen with the nature of the respective protein interfaces (analyzed by PDBePISA at <https://www.ebi.ac.uk/pdbe/pisa/>) and visualized in PyMOL with the InterfaceResidues plug-in (<https://pymolwiki.org/index.php/InterfaceResidues>), that shows TIE1 has a larger interface footprint than TIE2, and has a series of discriminating salt bridges and hydrogen bonds that are missing in TIE2.

A



B



Suppl. Figure 2

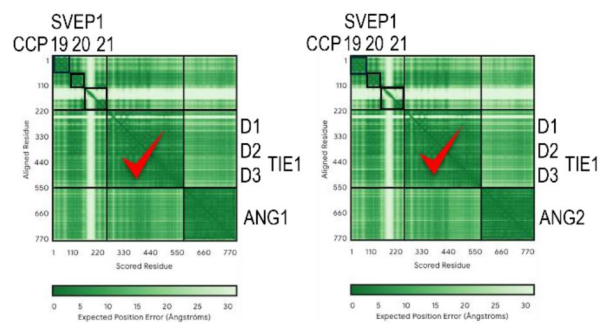
CCP5-EGF7 is sufficient, but not required for the lymphatic function of Svp1.

(A) Injection of different versions of murine Svp1 rescue constructs in *svep1* $-/-$ zebrafish zygotes at single cell stage was followed by quantifying the number of TD fragments at 5 dpf. For statistical analysis, the TD length of non-injected *svep1* mutants was compared to the injected mutants. Various constructs containing CCP5-EGF7 and CCP20 rescue the phenotype, with the CCP5-EGF7 fragment alone also being sufficient (wtSvp1: inj. n=9; uninj. n=3, PolC: inj. n=7; uninj. n=6, ΔN-EGFL6: inj. n=10; uninj. n=6, ΔN-PTX: inj. n=12; uninj. n=14, ΔN-PTXΔCCP21: inj. n=9; uninj. n=9, ΔN-PTXΔCCP22-C: inj. n=21; uninj. n=9, ΔN-CCP21: inj. n=9; uninj. n=7, ΔN-PTXΔ7-34: inj. n=22; uninj. n=21, CCP5-EGF7: inj. n=7; uninj. n=6, ΔN-EGFL7: inj. n=15; uninj. n=19). Mann-Whitney test was applied for statistical analysis. (B) Confocal images of the lymphatic vasculature (marked by a *ft4:mCitrine* transgene) in the zebrafish trunk of *svep1* sibling and mutants at 5 dpf. Siblings show complete TD, indicated by arrows. Non-injected *svep1* mutants show a loss of the TD, indicated by asterisks, while mutants injected with either CCP5-EGF7, ΔN-PTX, ΔN-PTXΔCCP21 or ΔN-PTXΔCCP22-C have significant more fragments of TD compared to the control. Scale bar = 100 μm. (C) Using CRISPR/Cas9 technology, an in-frame deletion of CCP5-EGF7 was generated in zebrafish. Homozygous mutants of 567 (CCP5-EGF7) show no lymphatic defects (*svep1* 567 $+/+$ n=22, *svep1* 567 $+/-$ n=53, *svep1* 567 $-/-$ n=21). Therefore, the region is not required for the lymphatic function of Svp1. Scale bar = 100 μm. TD, thoracic duct; dpf, days post fertilization

Suppl. Figure 3A

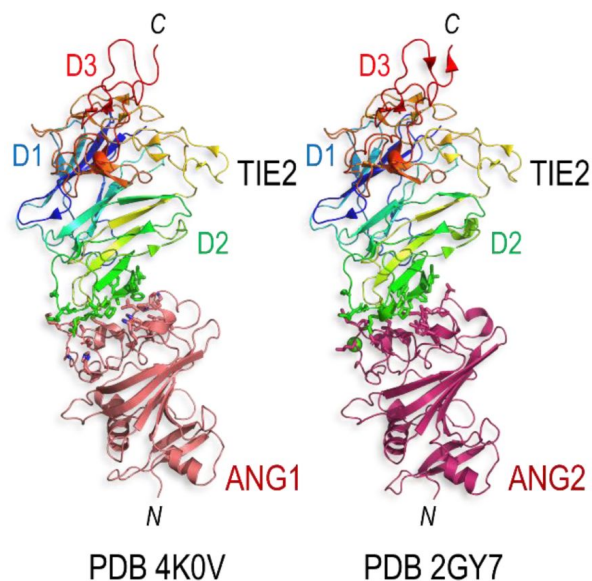
The PAE plots for the ternary assemblies of TIE1 D1-D3 bound to either ANG1 or ANG2 show equivalently strong confidence levels that improve the binding of just TIE1 to SVEP1. By contrast (not shown), TIE2 does not co-bind with ANG1/2 to SVEP1.

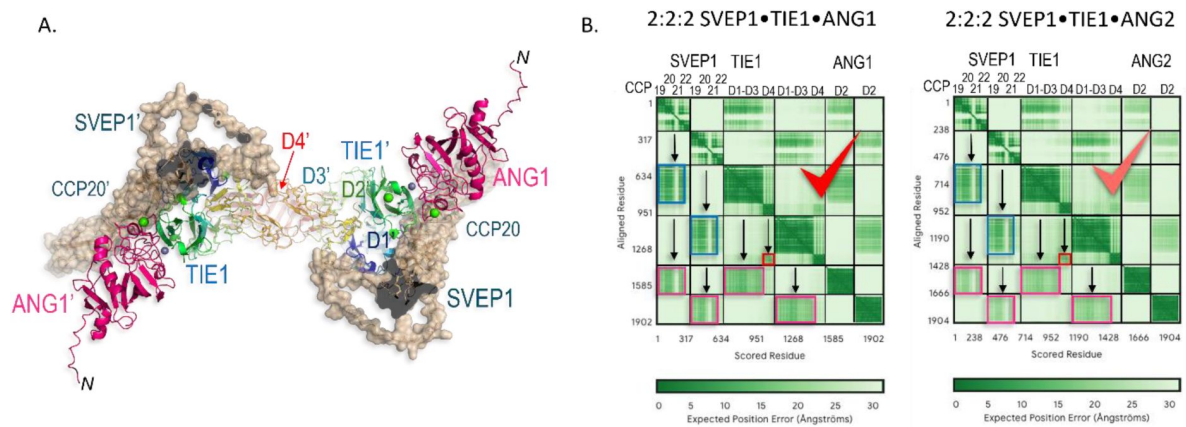
Comparative PAE plots



Suppl. Figure 3B

The conformation of the TIE1-ANG1/2 heteromers in the SVEP1-bound pose resembles the solved structures of TIE2 bound to either ANG1 or ANG2. However, the weaker binding of TIE2 to CCP19-21 of SVEP1 fails to nucleate the formation of the TIE2 ternary complex with SVEP1. PDE, predicted distance error.





Suppl Figure 5

(A) The top view of the model hexamer shows the side-by-side, antiparallel packing of TIE1 D3 domains that produces the domain-swapping of D4 Ig domains. (B) The PAE plot of the TIE1-ANG1-SVEP1 hexamer shows the many strong and confident interactions that build the assembly.v

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Reviewer #1 (Public review):

Summary:

In this manuscript, Uphoff et al. propose a structural and mechanistic model in which the multidomain ECM protein SVEP1 enables Angiopoietin (ANG) binding to the orphan receptor TIE1, thereby promoting downstream receptor phosphorylation and signaling. Using AlphaFold-based modeling, the authors predict that the CCP20 domain of SVEP1 binds to TIE1,

creating a composite surface that facilitates Angiopoietin association and TIE1 activation. The resulting ternary model (SVEP1-TIE1-ANG) offers a structural rationale for how SVEP1 converts TIE1 into a functional, ligand-responsive receptor. Additional models and biological assays suggest roles for other domains of SVEP1, such as CCP5-EGF-L7, although these interactions are predicted with low confidence. The authors interpret these findings as the first structural framework for how SVEP1 enables ANG-TIE1 signaling.

Strengths:

- (1) The central hypothesis - that SVEP1 enables ANG binding to the orphan receptor TIE1 - is biologically compelling and addresses an important question in vascular biology.
- (2) The AlphaFold-predicted ternary complex (SVEP1-TIE1-ANG) is plausible, high-confidence, and structurally consistent with prior functional data (e.g., poly-Ala scanning from Sato-Nishiuchi et al.).
- (3) The authors' model offers a potential explanation for the previously observed role of SVEP1 in enhancing ANG signaling through TIE1, and may represent the first structural insight into TIE1's transition from orphan to ligand-activated receptor.
- (4) The potential clinical implication - that a combinatorial ligand (ANG+SVEP1) can activate TIE1- could have translational relevance for vascular leak and inflammatory disease.

Weaknesses:

- (1) Lack of structural validation and mechanistic follow-up: Despite the promising AlphaFold model, there are no figures of the predicted interface, no residue-level interactions shown, no ipTM values reported, and no experimental follow-up to test the interface. PAE plots are incorrectly used as confidence justifications, which is not appropriate for complex predictions.
- (2) Biophysical validation is missing: No surface plasmon resonance (SPR), ITC, or biochemical assays are included to confirm ternary complex formation or quantify binding kinetics. Given the manuscript's structural focus, this is a major gap. For instance, an SPR experiment where ANG is immobilized, and TIE1 binding is measured {plus minus} SVEP1, would directly test the model. And allow direct comparison to ANG-TIE2.
- (3) Missed opportunity for mutagenesis-driven validation: The manuscript does not include any interface-targeted mutations, despite clear opportunities. For example, mutating T2595 in SVEP1 (to R) or mutating the TIE1-specific residues (residues PL 202-203 to LF) could strongly test the model and potentially reveal dominant-negative behaviors. E.g. A T2595 mutant should block ANG binding but not TIE1 binding.
- (4) Overinterpretation of weak models: The additional AlphaFold model involving the CCP5-EGFL7 domains binding TIE1 has extremely low confidence (ipTM < 0.15) when reexamined by this reader and should not be emphasized. There is no biophysical evidence or binding data (SPR) to support this interaction, and its inclusion detracts from the much stronger CCP20 model.
- (5) Language around modeling is overstated and potentially misleading: Terms like "unequivocal," "high-affinity," or "affirms strong binding" in reference to AlphaFold predictions are inappropriate. These are hypotheses -not confirmations - and must be tested at the biochemical level. This should be clarified throughout the manuscript to ensure non-experts do not misinterpret modeling confidence as binding affinity.
- (6) Negative stain EM data is not informative due to low resolution and lack of defined interfaces; unless replaced by higher-resolution Cryo-EM, this should be omitted. Better would be co-gel filtration, AUC, or SEC-MALLs with ANG-SVEP1-TIE1.

(7) Disjointed narrative: The manuscript presents a compelling mechanism involving CCP20-driven ANG binding to TIE1, but then becomes fragmented by introducing the low-confidence CCP5-EGFL7 model and speculative higher-order polymerization models that are not experimentally supported.

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Reviewer #2 (Public review):

Uphoff and colleagues present the results of a study focused on characterizing the binding of SVEP1 to TIE1 along with Angiopoietin-2. Starting with computational prediction of SVEP1 binding to TIE1, the authors identify the region of SVEP1 that serves as a high-affinity ligand for TIE1. Advanced studies identify a weak secondary binding site within SVEP1 that appears to be sufficient but not necessary for its interaction with TIE1 based on in vivo rescue experiments. The most novel contribution of the manuscript seems to be the identification of angiopoietin-1 and -2 as co-factors that seem to enhance the binding of SVEP1 with TIE1 and impact downstream AKT signaling. They propose a complex in which SVEP1 binds to TIE1 and ANG2.

Although the first set of results is essentially confirmatory, the identification of ANG-2 as a "co-factor" enhancing the binding of SVEP1 to TIE1 and associated downstream signaling (i.e., Figures 3 and 4) is novel and is of interest. However, the manuscript and its conclusions would greatly benefit from some clarifying details and additional experiments to ensure rigor and support specific claims.

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